IP Addressing and Subnetting

Q1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format

Class	First octet range	Default mask	Hosts per network	Real-world example
A	1 to 126	255.0.0.0 (/8)	16,777,214	Very large networks such as big ISPs, large companies and cloud data centres. Example: a company like Zomato's internal cloud network using the 10.0.0.0/8 private range.
B	128 to 191	255.255.0.0 (/16)	65,534	Medium to large networks such as a university or college campus network. Example: a college WiFi network using 172.16.0.0/16.
C	192 to 223	255.255.255.0 (/24)	254	Small networks such as homes and small offices. Example: a home WiFi router with the address 192.168.1.1 and devices like phones and laptops.
D	224 to 239	Not used (no network/host split)	Not applicable	Multicast, where one sender sends the same data to many receivers. Example: live IPTV or video streaming inside a network, and routing protocols such as OSPF using 224.0.0.5.
E	240 to 255	Not used	Not applicable	Reserved for experimental and research use. It is not used by normal devices or services.

Q2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Show your working steps for each calculation.

Given: IP address = 192.168.10.45, subnet mask = 255.255.255.0 (/24)

Step 1: Find the class and the network/host part

192 is in the range 192 to 223, so it is a Class C address. The mask 255.255.255.0 means the first 3 octets (24 bits) are the network part and the last octet (8 bits) is the host part.

Step 2: Network address (IP AND mask)

	Octets 1-3	Octet 4 (binary)	Octet 4 (decimal)
IP address	192.168.10	00101101	45
Subnet mask	255.255.255	00000000	0
AND result	192.168.10	00000000	0

Network address = 192.168.10.0

Step 3: Broadcast address (set all host bits to 1)

The host part is the last 8 bits. Setting all of them to 1 gives 11111111 = 255. So the broadcast address = 192.168.10.255.

Step 4: Range of valid host addresses

The first address (network address) and the last address (broadcast address) cannot be given to devices. First valid host = 192.168.10.0 + 1 = 192.168.10.1. Last valid host = 192.168.10.255 - 1 = 192.168.10.254.

Number of usable hosts = 2^8 - 2 = 256 - 2 = 254.

Final answer

Item	Value
Network address	192.168.10.0
Broadcast address	192.168.10.255
Valid host range	192.168.10.1 to 192.168.10.254
Usable hosts	254

Q3. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.Calculation

- Subnets needed = 4. Bits to borrow n: 2^n >= 4, so n = 2 (2^2 = 4).
- New prefix = 24 + 2 = /26. Subnet mask = 255.255.255.192 (last octet 11000000 = 192).
- Block size = 256 - 192 = 64 addresses per subnet. Usable hosts per subnet = 2^6 - 2 = 62.

Group	Network address	Subnet mask	Valid host range	Broadcast address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 to 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 to 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 to 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 to 10.0.0.254	10.0.0.255

Q4. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.Show your calculation steps and final answers.

Step 1: Bits needed for subnets

We need at least 20 subnets. Using 2^n >= 20: 2^4 = 16 (too few) and 2^5 = 32 (enough). So we borrow n = 5 bits from the host part.

Step 2: New subnet mask

- Original prefix = /16, new prefix = 16 + 5 = /21.
- The borrowed 5 bits are in the third octet: 11111000 = 128 + 64 + 32 + 16 + 8 = 248.
- New subnet mask = 255.255.248.0 (/21).

Step 3: Hosts per subnet

- Host bits = 32 - 21 = 11.
- Total addresses = 2^11 = 2048. Usable hosts = 2048 - 2 = 2046.

Step 4: Subnet details

- Block size in the third octet = 256 - 248 = 8. So subnets increase by 8: 172.16.0.0, 172.16.8.0, 172.16.16.0, and so on up to 172.16.248.0.

- Total subnets = 2^5 = 32 (more than the 20 needed, so 12 are free for future buildings).

Subnet	Network address	Valid host range	Broadcast address
1	172.16.0.0/21	172.16.0.1 to 172.16.7.254	172.16.7.255
2	172.16.8.0/21	172.16.8.1 to 172.16.15.254	172.16.15.255
3	172.16.16.0/21	172.16.16.1 to 172.16.23.254	172.16.23.255
32	172.16.248.0/21	172.16.248.1 to 172.16.255.254	172.16.255.255

Final answer

Item	Value
New subnet mask	255.255.248.0 (/21)
Number of subnets	32 (at least 20 required)
Hosts available per subnet	2046

Q5 A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

Error: The address 192.168.1.300 is not a valid IPv4 address.

An IPv4 address has four parts (octets), and each octet is 8 bits, so each value can only be from 0 to 255. In 192.168.1.300 the last octet is 300, which is more than 255. Because the address is invalid, the device cannot get on the network and cannot connect.

What a valid address looks like

- The first three octets must match the network, 192.168.1, and the last octet must be between 1 and 254 (with the usual mask 255.255.255.0).
- Valid examples: 192.168.1.30 or 192.168.1.200. The numbers 0 and 255 cannot be used, as they are the network and broadcast addresses.
- Do not use the router's own address (usually 192.168.1.1) or an address already used by another device, as that causes a conflict.

Suggestion for the friend: Set the IP to automatic (DHCP) in the network settings so that the router gives a correct address, or type a valid address manually, such as 192.168.1.30 with subnet mask 255.255.255.0 and gateway 192.168.1.1.